

SeismoLink

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01 The Problem

What case study/problem
we are working on





THE PROBLEM

- We decided to focus on case study #3
- Address the absence of accessible earthquake alert infrastructure, particularly for underserved communities in Haiti
- The lack of information about incoming earthquakes especially during the absence of power can severely effect the safety and healthcare of individuals

WHY HAITI?

- One of the poorest and most underdeveloped countries with challenges in communication infrastructure
- Ranks 2nd globally in earthquake-related fatalities [1]
- For example, the 2010 earthquake caused over **230,000 deaths** [2]
- Although Haiti became an authorized user of the International Charter on Space and Major Disasters in 2019 it still faces limitations in informing citizens of natural disasters [3]
- The country has unreliable electricity, limited mobile coverage and unable to deliver critical warning information to underserved communities [4]
- There is also a lack of operational seismic monitoring stations
- Inability to deliver critical warning information to remote and underserved communities cases high mortality rates



Stakeholders

- Local Disaster Management Agencies
- NGOs / Humanitarian Organizations
- Satellite Data Providers / Alert Broadcasters





02 The Solution

Our initial plan to solve
the problem





- The system functions as an earthquake detection and alarm unit, specifically designed for accessibility and sustainability
- It operates independently of the electrical grid by utilizing solar panels as a renewable energy source
- This prevents communities from relying on inadequate power sources
- To ensure effective communication across diverse user needs, the device will provide alerts through both visual and audio outputs
- Allows for those with disabilities or accessible needs to be informed ahead of time to prepare or evacuate



THE SOLUTION

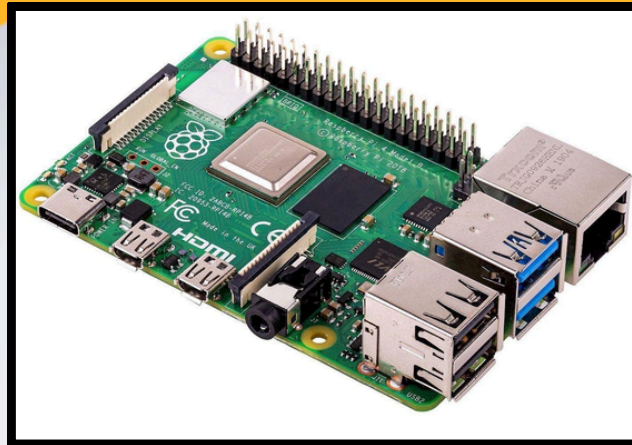


03 The Proposition

The project details



HARDWARE COMPONENT



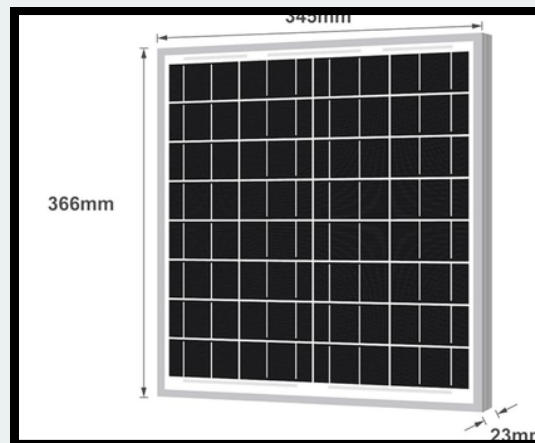
**32GB Raspberry Pi
microSD Card**



Satellite Modem



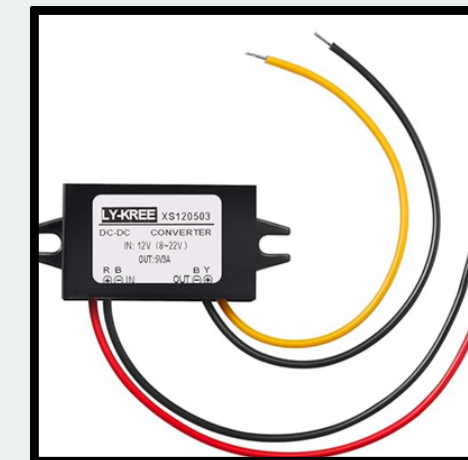
Output Devices



5-10 W Solar Panel



**12V 7Ah Sealed Lead
Acid Battery**

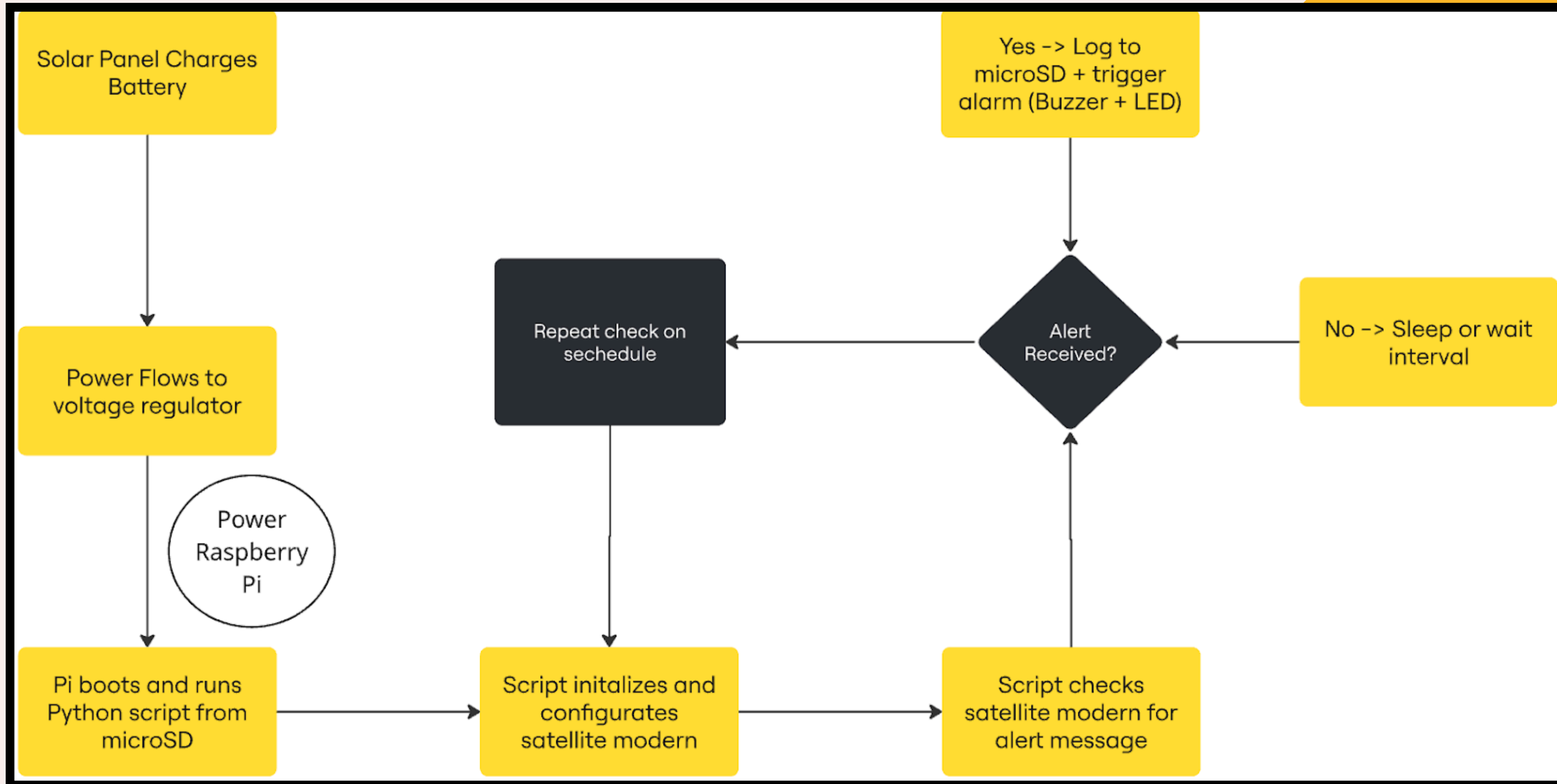


**12V to 5V buck
converter**

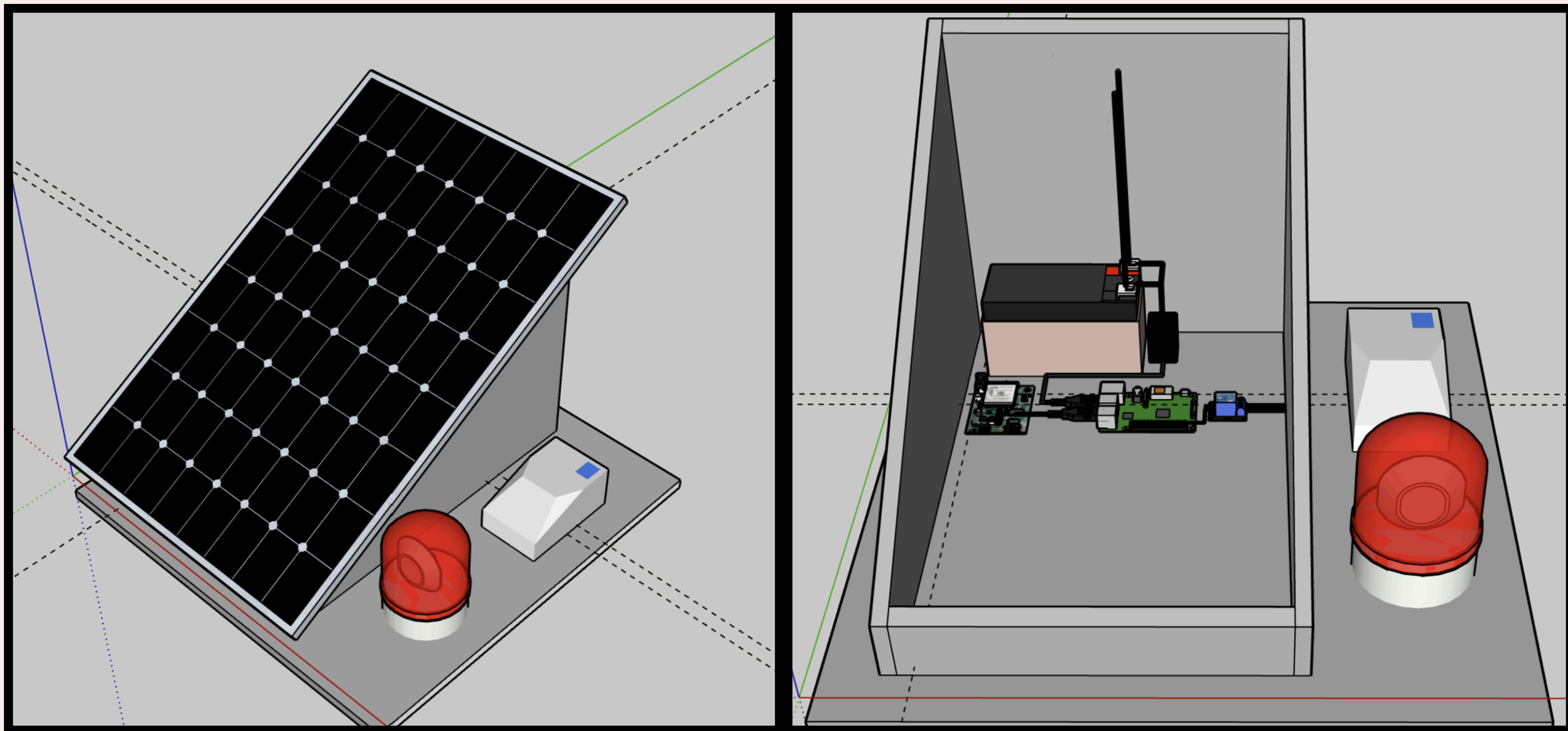
COST ESTIMATION

Component	Cost
Raspberry Pi	\$80
Buzzer	\$10
Strobe Led light	\$10
5-10W solar panel	\$40
Power supply	\$30
32GB Raspberry Pi microSD card	\$20
Satellite modem	\$430
Total	\$620
Total After HST	\$700

Flow Chart

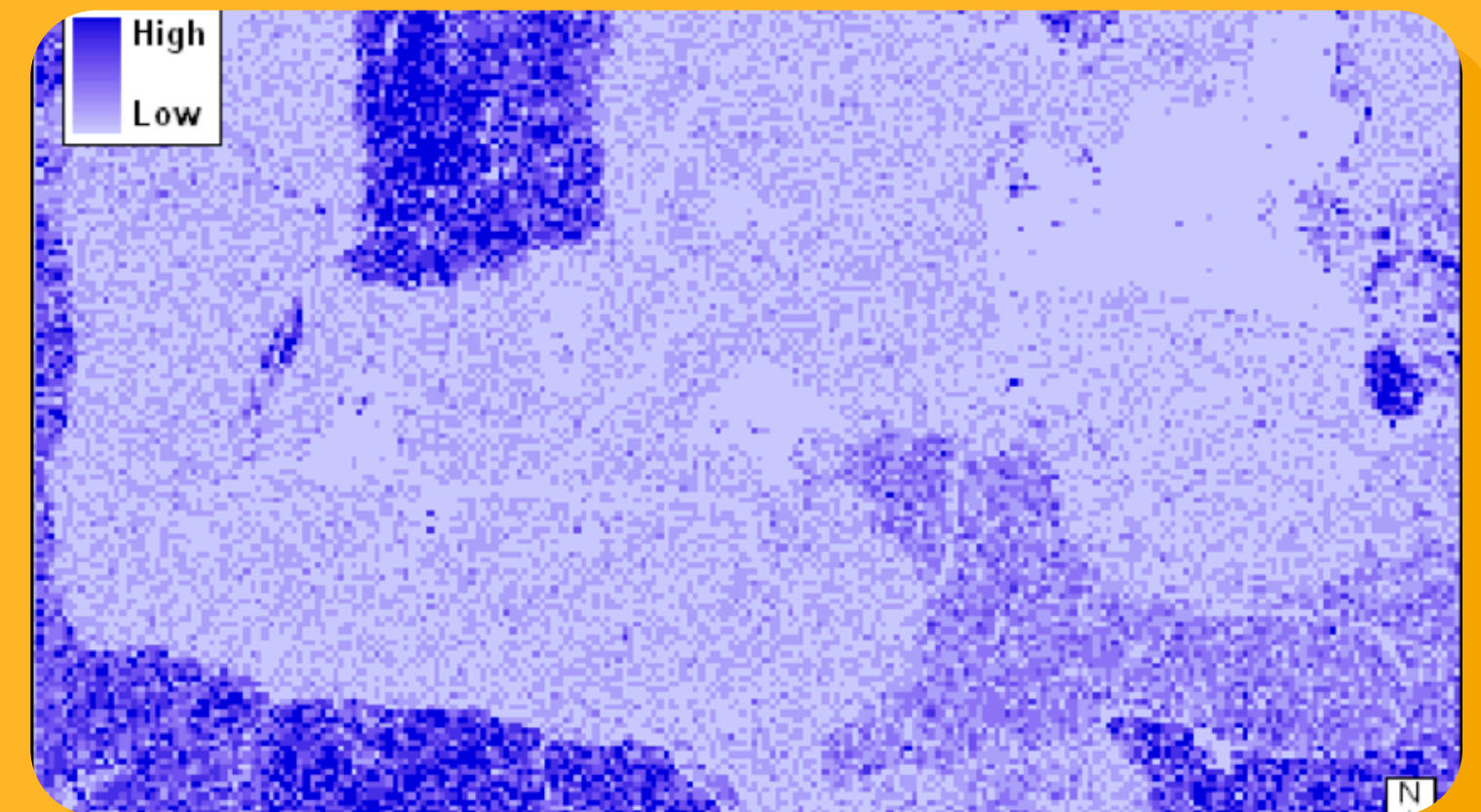
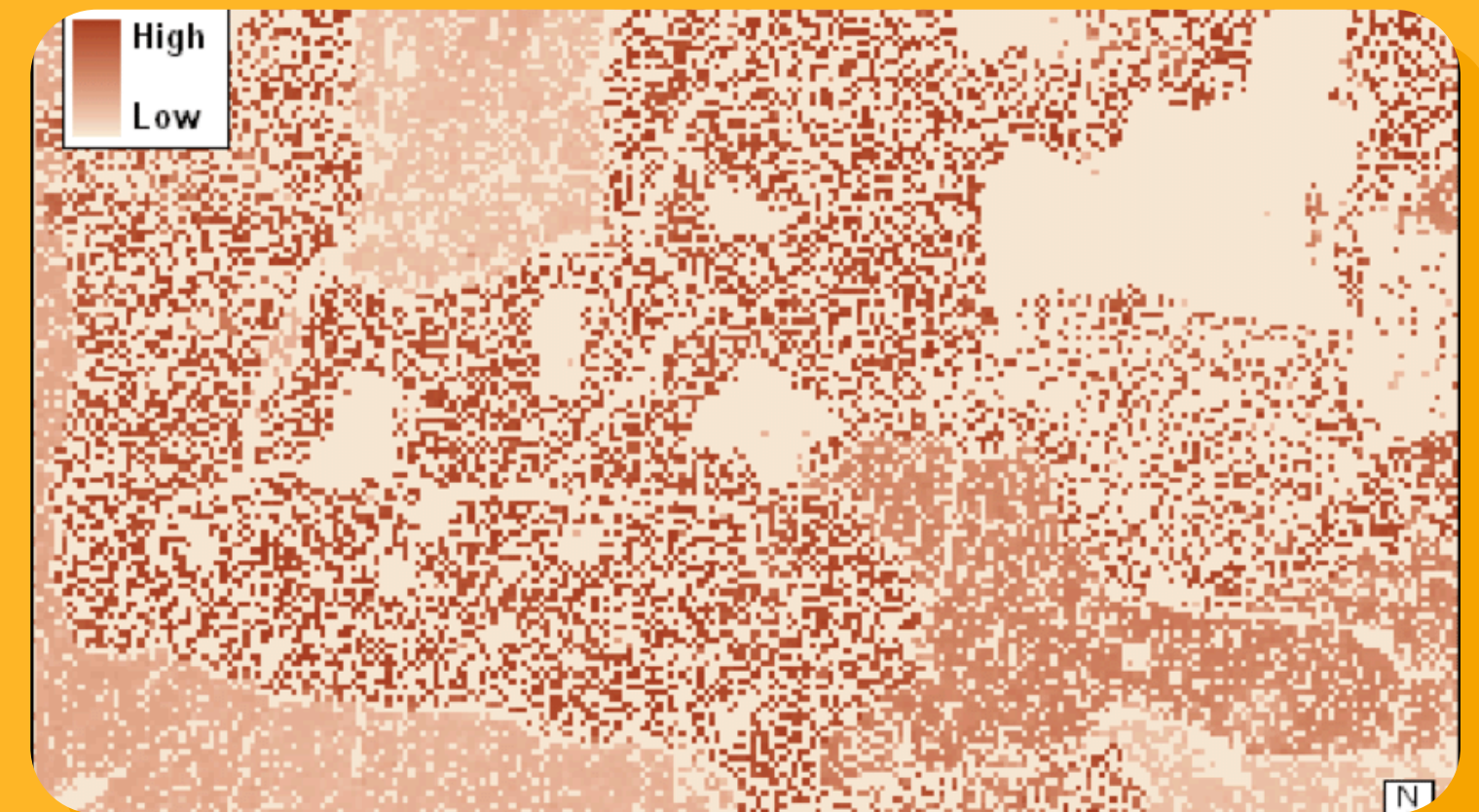


3D CAD MODEL



THE SOFTWARE

1. Input Image Acquisition:
 - a. Load two GeoTIFF files as inputs:
 - i. Image A: Ground displacement data recorded ***before*** the seismic event
 - ii. Image B: Ground displacement data recorded ***after*** the seismic event
2. Displacement Calculation:
 - a. Perform pixel-wise subtraction between Image B and Image A: $\Delta D = D(\text{after}) - D(\text{before})$
 - b. Result: A differential displacement map indicating the magnitude of ground movement at each pixel.
3. Thresholding for Physical Change:
 - a. Apply a threshold of 10 mm (1 cm) to the filtered displacement data.[5]



THE SOFTWARE

4. : Pixel Count Analysis

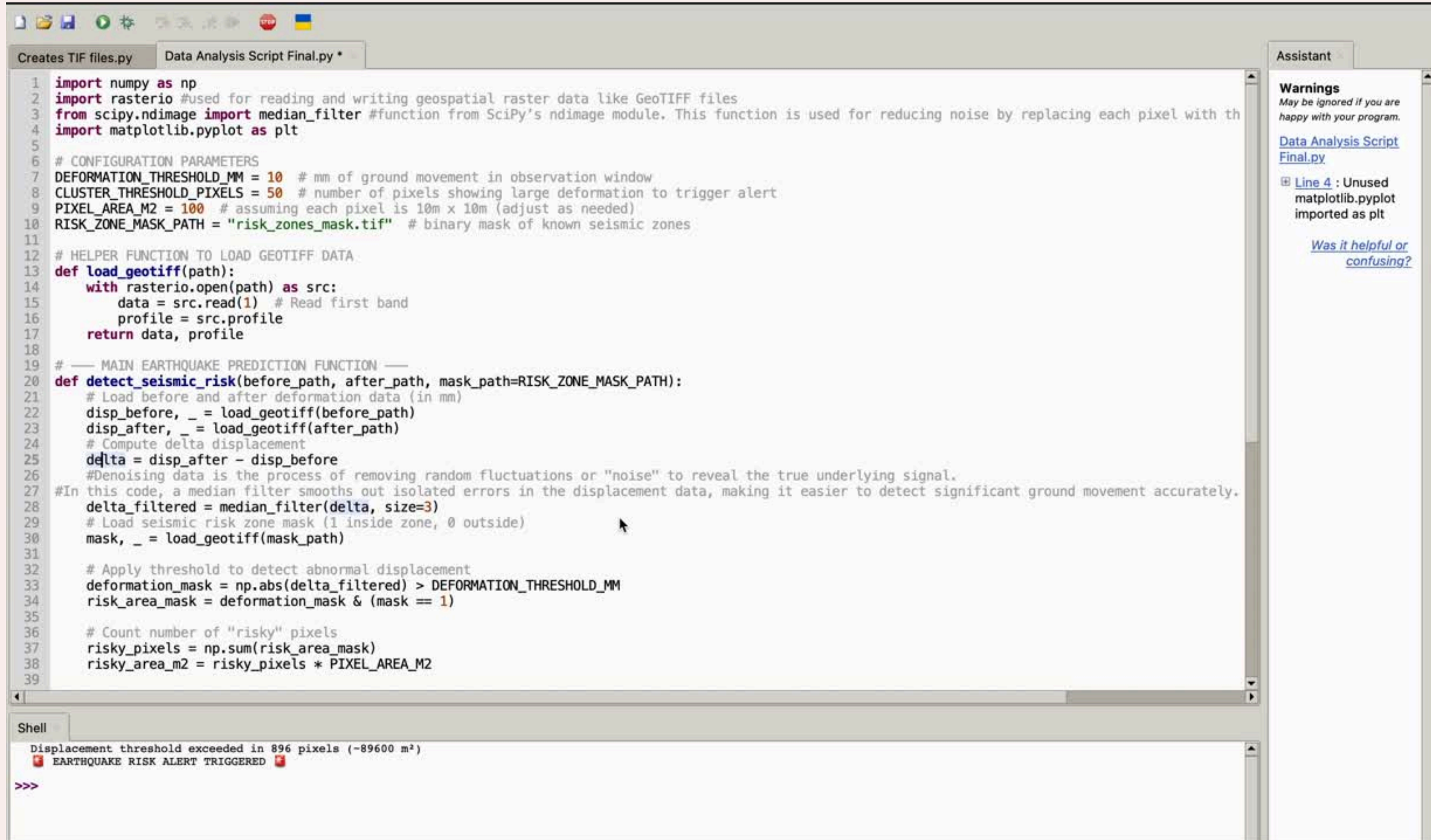
- a. Count the number of pixels within the risk zones where the displacement exceeds the 10 mm threshold

5. Alert Triggering Mechanism:

- a. If the number of significantly displaced pixels exceeds 50, trigger an earthquake alert.



DEMO



```
1 import numpy as np
2 import rasterio #used for reading and writing geospatial raster data like GeoTIFF files
3 from scipy.ndimage import median_filter #function from SciPy's ndimage module. This function is used for reducing noise by replacing each pixel with th
4 import matplotlib.pyplot as plt
5
6 # CONFIGURATION PARAMETERS
7 DEFORMATION_THRESHOLD_MM = 10 # mm of ground movement in observation window
8 CLUSTER_THRESHOLD_PIXELS = 50 # number of pixels showing large deformation to trigger alert
9 PIXEL_AREA_M2 = 100 # assuming each pixel is 10m x 10m (adjust as needed)
10 RISK_ZONE_MASK_PATH = "risk_zones_mask.tif" # binary mask of known seismic zones
11
12 # HELPER FUNCTION TO LOAD GEOTIFF DATA
13 def load_geotiff(path):
14     with rasterio.open(path) as src:
15         data = src.read(1) # Read first band
16         profile = src.profile
17     return data, profile
18
19 # — MAIN EARTHQUAKE PREDICTION FUNCTION —
20 def detect_seismic_risk(before_path, after_path, mask_path=RISK_ZONE_MASK_PATH):
21     # Load before and after deformation data (in mm)
22     disp_before, _ = load_geotiff(before_path)
23     disp_after, _ = load_geotiff(after_path)
24     # Compute delta displacement
25     delta = disp_after - disp_before
26     #Denoising data is the process of removing random fluctuations or "noise" to reveal the true underlying signal.
27     #In this code, a median filter smooths out isolated errors in the displacement data, making it easier to detect significant ground movement accurately.
28     delta_filtered = median_filter(delta, size=3)
29     # Load seismic risk zone mask (1 inside zone, 0 outside)
30     mask, _ = load_geotiff(mask_path)
31
32     # Apply threshold to detect abnormal displacement
33     deformation_mask = np.abs(delta_filtered) > DEFORMATION_THRESHOLD_MM
34     risk_area_mask = deformation_mask & (mask == 1)
35
36     # Count number of "risky" pixels
37     risky_pixels = np.sum(risk_area_mask)
38     risky_area_m2 = risky_pixels * PIXEL_AREA_M2
39
```

Displacement threshold exceeded in 896 pixels (-89600 m²)
🚨 EARTHQUAKE RISK ALERT TRIGGERED 🚨

>>>

Assistant

Warnings
May be ignored if you are happy with your program.

[Data Analysis Script Final.py](#)

ⓧ Line 4 : Unused matplotlib.pyplot imported as plt

[Was it helpful or confusing?](#)

CHALLENGES

- A key difficulty in the design process was figuring out possible conversion of satellite ground movement images to GEOTIFF files
- We needed this process to occur without an internet connection
- Estimating how often we would take images from the satellite to monitor ground movement



THE FUTURE TIMELINE



1. Disasters

Implement device for other natural disasters



2. Countries

Include other countries such as Indonesia & Papua New Guinea



3. Communication with EMS

Sends a distress signal to government services with GPS tracking



4. Communication between Devices

Communicate with other nearby devices to ask for help

THANK YOU!

Feel free to ask any questions!



CREDITS

[1] - https://en.wikipedia.org/wiki/Lists_of_21st-century_earthquakes

[2] - https://en.wikipedia.org/wiki/2010_Haiti_earthquake

[3] - <https://www.innovations-report.com/agriculture-environment/earth-sciences/satellite-map-haiti-earthquake-146579/#:~:text=To%20meet%20the%20requirements%20of,2%20and%20Envisat>

[4] - https://internews.org/wp-content/uploads/legacy/resources/IAA_Haiti_2010_0.pdf

[5] - <https://sentinel.esa.int/web/sentinel/missions/sentinel-1>

